Swiss Institute of Banking and Finance
University of St.Gallen

INDEPENDENT STUDIES

SECOND SESSION

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3,130 Corporate Finance

Major in Business Administration (BWL)

5. Cost of Capital (BMAE, Chapters 4, 5, 6, and 9)

5.1. True / False Questions

a. The company cost of capital is defined as the expected return on a portfolio of all the company's outstanding debt and equity securities.

   a. True
   b. False

b. The weighted average cost of capital is the discount rate for the firm's average-risk projects.

   a. True
   b. False

c. When estimating the weighted-average cost of capital, debt and equity enter at market value.

   a. True
   b. False

d. The market risk premium is the difference between the return on the market and the risk-free interest rate.

   a. True
   b. False

e. Each project, a company seeks to invest into, should be discounted by its own discount rate which considers the project's appropriate risk.

   a. True
   b. False

f. "Pure plays" are firms in the same line of business as the project considered, not involved in any other type of business, and privately held.

   a. True
   b. False

g. Projects with long-term cash flows are less sensitive to changes in the risk-free rate or market risk premium.

   a. True
   b. False

h. Unbiased forecasts of project's cash flows incorporate all possible outcome, including those that are specific to the project and those that stem from economywide events.

   a. True
   b. False

i. The company cost of capital is the correct discount rate for any project a company considers, because high risks of some projects are offset by the low risk of other projects.

   a. True
   b. False

j. Interest payments and dividends are tax deductible.

   a. True
   b. False

k. WACC is based on a firm's future characteristics.

   a. True
   b. False

l. When using the WACC one assumes that risks remain unchanged for the life of the project.

   a. True
   b. False

m. One key assumption of the APV approach is that the firm actively rebalances in order to maintain its debt ratio.

   a. True
   b. False

n. The APV method is most useful when side-costs are numerous and important.

   a. True
   b. False

o. There can never be more than one value of the IRR for any sequence of cash flows.

   a. True
   b. False

p. The internal rate of return is the discount rate that makes the NPV of a project's cash flows equal to zero.

   a. True
   b. False

q. Depreciation expense acts as a tax shield in reducing taxes.

   a. True
   b. False

r. According to the APV approach, the value of the firm is the sum of the base-case NPV of the firm if it had no debt, the present value of the tax shield, and the present value of all other side effects.

   a. True
   b. False

s. The spread of junk bond yields over U.S. Treasury yields is generally lower than the spread of investment-grade bonds over U.S. Treasury yields.

   a. True
   b. False

t. Bonds rated Ba/ BB or below are investment grade.

   a. True
   b. False

u. Risky bonds have a positive beta and thus have market risk. This explains why there are time when firm defaults are more likely.

   a. True
   b. False

5.2. Cost of equity

The X-Corporation reports earnings per share (EPS) of 4.5 EUR, whereof 68% are distributed to the shareholders. The beta of X-Corporation is 0.78, the risk-free interest rate is 4% and the market return is 8.5%. What is the most likely cost of equity capital for the X-Corporation?

a. 3.51%
b. 5.50%
c. 7.51%
d. 8.50%
e. 10.63%

5.3. Asset beta

The capital structure of ABC Corp consists of 50% long-term bonds and 50% common equity. The debt securities have a beta of 0.1. The company's equity beta is 1.2. What is the asset beta?

a. 0.10
b. 0.60
c. 0.65
d. 0.70
e. 1.20

5.4. WACC

Shares of Olma Inc. trade at USD 40 per share. There are CHF 3,000,000 shares outstanding and the expected rate of return is 18%. Olma Inc. has issued bonds with a book value of CHF 75,000,000. The debt is trading at a discount of 10% to its book value and has a yield to maturity of 9%. Olma's (marginal) tax rate is 40%. What is Olma's after-tax WACC?

a. 10.11%
b. 10.15%
c. 13.15%
d. 13.46%
e. 14.76%

5.5. Project costs of capital

Johnson Inc. has two divisions. The first division manufactures consumer staples such as antiseptic soaps, detergents for dish washers, washing machines, and power cleansers. The asset beta of firms that only produce sanitary articles is 0.85. The second division manufactures investment goods for the household, such as fridges, refrigerators, and smaller appliances like flat irons. The average asset beta of comparable pure-play firms is 1.2. The first division accounts for 40%, the second for 60% of Johnson's market value. The company has no debt. The risk-free rate is 2%, and the market risk premium ($r_m - r_f$) is 7.3%.

Suppose the company has two projects: project A in the consumer staples division with an IRR of 9% and project B in the investment goods division with an IRR of 10%.
Which project should be taken (assuming that both projects have downward sloping NPV functions so that the simple IRR rule works)?

a. A
b. B
c. Both projects
d. None of the both projects.
e. None of the answers a. to d. are correct.

5.6. Net present value rule

You are given a job to make a decision on project portfolio X, which is composed of three independent projects A, B, and C that have NPVs of + USD 70, -USD 40, and + USD 100, respectively. How would you go about making the decision about whether to accept or reject the project?

a. Accept the project portfolio X as it has a positive NPV.
b. Reject the project portfolio X.
c. Break up the project into its components: accept A and C, but reject B.
d. Break up the project into its components: accept C.
e. None of the answers a. to d. are correct.

5.7. Firm's cost of capital

The total market value of the common stock of Flower Inc. is USD 6 million, and the total value of its debt is USD 4 million. The beta of Flower's stock is currently 1.3 and the expected risk premium on the market is 8%. The Treasury bill rate is at 2%. Assume for simplicity that Flower's debt is risk-free and no taxes have to be paid.

5.7.1. Cost of equity

What is the most likely required return on Flower's stock?

a. 9.4%
b. 10.4%
c. 11.4%
d. 12.4%
e. 13.4%

5.7.2. Cost of capital

What is the most likely company cost of capital?

a. 4.47%
b. 8.08%
c. 8.24%
d. 10.64%
e. 12.44%

5.7.3. Cost of capital

What is the most likely discount rate for an expansion of the company's average business?

a. 4.47%
b. 6.25%
c. 8.24%
d. 12.40%
e. Cannot say.

5.7.4. After-tax WACC

Suppose that the government raises taxes so that Flower's tax rate is 20%. What is the most likely company cost of capital after taxes?

a. 4.47%
b. 8.08%
c. 8.24%
d. 10.64%
e. 12.44%

5.8. Firm's asset beta

Cold Inc. has the following capital structure:

Security	Beta	Market Value (USD million)
Debt	0	100
Preferred stock	0.2	40
Common stock	1.2	300

What is Cold's most likely asset beta?

a. 0.75
b. 0.82
c. 0.84
d. 0.90
e. 1.08

5.9. Expected cash flows

A drilling company is exploring new territory. It plans to drill a series of new wells close to a producing oil field. According to firm estimates, around 15% of the new wells will be dry holes. However, even if a new well strikes oil, there is still uncertainty about the production: 30% of new wells that strike oil produce only 1,000 barrels a day; 70% produce 5,000 barrels per day. The future oil price is expected to be USD 100 per barrel. What are the most likely annual cash revenues from the new perimeter well, assuming that oil production is possible on 365 days per year?

a. USD 99,280,000
b. USD 105,775,000
c. USD 110,225,000
d. USD 117,895,000
e. USD 138,700,000

5.10. After-tax WACC

A company is 20% financed by risk-free debt. The risk-free interest rate is 10%, the expected market risk premium is 7.1%, and the beta of the company's common stock is 0.6.

5.10.1.

What is the most likely company cost of capital?

a. 11.2%
b. 12.9%
c. 13.4%
d. 14.3%
e. 15.4%

5.10.2.

What is the most likely company after-tax WACC, assuming a tax rate of 40%?

a. 10.43%
b. 12.61%
c. 13.98%
d. 14.44%
e. 15.10%

5.11. Net present value

Music Company is considering investing into a new project. The project will need an initial investment of USD 2,400,000 and will generate USD 1,200,000 (after-tax) cash flows per year for the next three years. What is the most likely NPV for the project if the after-tax cost of capital is 15%?

a. USD 123,846
b. USD 169,935
c. USD 339,870
d. USD 600,000
e. USD 1,200,000

5.12. Project valuation

A project requires an investment of USD 900 today. It generates sales of USD 1,100 per year forever but can also be terminated each year. Costs are USD 600 for the first year and will increase by 20% per year. Assume all sales and costs occur at year-end and ignore taxes. What is, most likely, the highest possible NPV of the project at a 12% discount rate?

a. USD 57.51
b. USD 65.00
c. USD 71.27
d. USD 100.00
e. Cannot be calculated as $g > r$.

5.13. Project valuation

The B-Corporation thinks about expanding its business. For this reason, the B-Corporation will set up a subsidiary. The initial costs are CHF 1,500 and the operating costs will be CHF 200 each year. The expected cash inflows (in CHF) are 150 (year 1), 900 (year 2), 1,100 (year 3), 1,200 (year 4). The discount rate is 8%.

5.13.1.

What is the most likely NPV of this project?

a. CHF 503.32
b. CHF 595.91
c. CHF 613.21
d. CHF 1,165.75
e. CHF 2,003.32

5.13.2.

As an alternative to the establishment of a subsidiary, the B-Corporation has the possibility to use the network of a partner company. In this case, the same cash inflows can be achieved but royalty fees of CHF 650 have to be paid each year.

Which statement is correct?

a. The establishment of the subsidiary should be preferred to the use of the network of the partner company.
b. The use of the network of the partner company should be preferred to the establishment of the subsidiary.
c. The company is indifferent as both the subsidiary and the network of the partner company result in the same NPV.
d. It is not possible to take a decision on the basis of these data.
e. None of the answers a. to d. are correct.

5.14. APV

Tomato Inc. shows the following financial characteristics. The firm's market value of assets is CHF 5,000. The market value of debt is CHF 2,000. The returns on equity and debt are 12% and 4%, respectively. Tomato Inc. pays a marginal corporate tax rate of 35%. The firm considers a new project with perpetual pre-tax cash flows of CHF 66, an initial investment of CHF 210, and the same risk structure as the firm.

5.14.1. Base-case

What is the project's most likely base-case (i.e., all equity) NPV?

a. CHF 210.00
b. USD 277.50
c. CHF 310.63
d. CHF 540.00
e. CHF 590.97

5.14.2. Constant debt ratio

What is the value of the project holding the debt ratio constant over time?

a. CHF 211.45
b. CHF 237.01
c. CHF 254.12
d. CHF 272.52
e. CHF 290.86

5.14.3. Fixed amount of debt

What is the value of the project holding the amount debt constant at CHF 84 over time?

a. CHF 155.23
b. CHF 208.30
c. CHF 253.72
d. CHF 306.90
e. CHF 357.34

5.15. APV

A project has forecasted cash flows of USD 110 in year 1 and USD 121 in year 2. The interest rate is 5%, the estimated risk premium on the market is 10%, and the project has a beta of 0.5.

Using a constant risk-adjusted discount rate, what is the ratio of the certainty-equivalent cash flows to the expected cash flows in year 2?

a. 0.8326
b. 0.8732
c. 0.9112
d. 0.9545
e. 0.9972

6. Corporate Financing and the Issuance of securities (BMAE, Chapters 13 and 14)

6.1. True / False Questions

a. Most capital invested by U.S. companies is funded by retained earnings and cash flows allocated to depreciation.

   a. True
   b. False

b. Historical debt ratios, i.e. in 1996 and 1997, were lower in the United States than in many European countries, i.e. Germany, Italy, France, and Switzerland.

   a. True
   b. False

c. Net stock issues by U.S. nonfinancial corporations between 1991 and 2017 are small but positive in most years.

   a. True
   b. False

d. In the United States, the majority of common stocks were owned by individual investors in 2017.

   a. True
   b. False

e. Hedge funds provide unsophisticated retail investors with low-cost diversification.

   a. True
   b. False

f. The sale of insurance policies is one source of financing for insurance companies.

   a. True
   b. False

g. Exchange traded funds are hedge funds that can be bought and sold on the stock exchange.

   a. True
   b. False

h. Rather than investing directly in corporate equities, most households prefer to pool their risk in a hedge fund.

   a. True
   b. False

i. Insurance companies are massive investors in corporate debt which are especially important for long-term financing of businesses.

   a. True
   b. False

j. Banks are massive investors in corporate equity.

   a. True
   b. False

k. The defined-contribution plan is the most common type of pension plan where the pension pot depends on the rate of return earned on the contributions by the employer and employee to a pension fund.

   a. True
   b. False

l. Venture Capital companies generally advance the money in stages.

   a. True
   b. False

m. Venture Capital companies typically provide mentorship and connections for the entrepreneurs, and also often place a representative on the board of the firm.

   a. True
   b. False

n. Some young companies grow with the aid of equity investment provided by wealthy individuals known as angel investors.

   a. True
   b. False

o. Venture capitalist provide first-stage financing sufficient to cover all development expenses. Second-stage financing is provided by stock issued in an IPO.

   a. True
   b. False

p. Underpricing in an IPO is only a problem if the original investors are selling part of their holdings.

   a. True
   b. False

q. Stock prices on average fall when companies announce a new equity issue. This may not be attributable to the extra supply of stocks but simply to information released by the decision to issue.

   a. True
   b. False

r. At the time of the announcement of a rights issue, the rights issue will give the shareholder the opportunity to buy one new share for less than the market price.

   a. True
   b. False

6.2. Preferred stocks

In 2018, Beta Corporation earned gross profits of $760,000.

6.2.1.

Suppose that Beta was financed by a combination of common stocks and $1 million of debt. The interest rate on the debt was 10%, and the corporate tax rate in 2018 was 21%. How much profit was available for common stockholders after payment of interest and corporate taxes?

a. $517,400
b. $518,400
c. $519,400
d. $520,400
e. $521,400

6.2.2.

Now suppose that instead of issuing debt, Beta was financed by a combination of common stock and $1 million of preferred stock. The dividend yield on the preferred stock was 8%, and the corporate tax rate was still 21%. How much profit was available for common stockholders after payment of preferred dividends and corporate taxes?

a. $517,400
b. $518,400
c. $520,400
d. $521,400
e. $522,400

6.3. Venture Capital

The Bionic Company is a young software company owned by 2 entrepreneurs. It currently needs to raise $400,000 to support its expansion plans. A venture capitalist is willing to provide the cash in return for a 40% holding in the company. Under the plans for the investment, the venture capital company will hold 10,000 shares in the Bionic Company, and the 2 entrepreneurs will have combined holdings of 15,000 shares.

6.3.1.

What is the total after-the-money valuation of the firm?

a. $600,000
b. $1,000,000
c. $1,500,000
d. $2,100,000
e. $2,450,000

6.3.2.

What value is the venture capitalist placing on each share?

a. $40
b. $66
c. $87
d. $100
e. $112

6.4. Issue costs

For each of the following issues, which is likely to involve the lower proportionate underwriting and administrative costs: I) a large issue; II) a small issue; III) a bond issue; IV) a common stock issue; V) Initial Public Offering; VI) subsequent issue of stock.

a. I, IV and V only
b. I, III and VI only
c. I, III and V only
d. II, III and VI only
e. II, IV and V only

6.5. Type of seasoned issue

In the following, four methods of security issues are listed. For each issue method, two alternative securities are provided. Please choose for each issue method the security that is more likely to be issued using this method.

- Rights issue: I) initial public offer or II) further sale of an already publicly traded stocks.
- Rule 144A issue: III) international bond issue or IV) U.S. bond issue by a foreign corporation.
- Private placement: V) issue of existing stock or VI) bond issue by an industrial company.
- Shelf registration: VII) initial public offer or VIII) bond issue by a large industrial company.

a. I, III, V and VII
b. I, IV, VI, and VIII
c. II, IV, VI and VIII
d. II, III, V and VII
e. II, IV, V and VII

6.6. Sarbanes-Oxley Act

Which statement is false about of the consequences of Sarbanes-Oxley Act?

a. Fewer high-growth entrepreneurial companies are going public and more of them opt to provide liquidity and an exit for investors by selling out to larger companies.
b. Sarbanes-Oxley Act was clearly the main reason for 2008 and 2009 experiencing a smaller percentage of venture-backed IPOs than any other year since 1985, this decline being concentrated among large venture-backed firms.
c. As job growth accelerates when companies go public and decelerates when companies are acquired, Sarbanes-Oxley Act is responsible for hurting job creation.
d. Sarbanes-Oxley Act has deterred U.S. companies from going public in New York and has rather induced them to list in London.
e. The number of public corporations has fallen by about 50% since its high in 1996, as the incentive for remaining private under Sarbanes-Oxley Act is stronger than it previously was.

6.7. Issue costs, underpricing and money left on the table

The Alpha Company offered 100 shares for sale in an IPO. Half of the shares were sold by the company and the other half by existing shareholders, each of whom sold exactly half of their existing holding. The offering price to the public was $50, and the underwriters received a spread of 7%. The issue was heavily oversubscribed, and by the end of the first day of trading, the stock price rose to $160.

6.7.1

What were the proceeds of the issue to the company?

a. $1,636
b. $1,825
c. $2,136
d. $2,325
e. $2,546

6.7.2

What were the proceeds of the issue to the shareholders?

a. $1,436
b. $1,625
c. $1,836
d. $2,146
e. $2,325

6.7.3

How much commission did the underwriters receive?

a. $350
b. $377
c. $380
d. $417
e. $440

6.7.4

How much money was left on the table?

a. $5,000
b. $8,000
c. $11,000
d. $14,000
e. $17,000

6.7.5

What was the cost of the underpricing to the selling shareholders?

$5,675
$7,755
$9,945
$11,135
$13,323

6.8. Prospectus for an IPO

Marvin Enterprises shows the following information in its prospectus for its IPO. Marvin has 900,000 shares of common stocks outstanding. The company holds 500,000 shares which are being sold directly by the Marvin Company. The other 400,000 shares are being sold by investors already holding shares of Marvin (Selling Stockholders). The Marvin Enterprises will not receive any of the proceeds from the sale of shares by the Selling Stockholders. Before this offering, there has been no public market for the common stock.

	Price to public	Underwriting discount	Proceeds to company	Proceeds to Selling Stockholders
Per share	$80.00	$5.60	$74.40	$74.40
Total	$72,000,000	$5,040,000	$37,200,000	$29,760,000

The proceeds to company are computed before deducting administrative fees payable by the company estimated at $820,000, of which $455,555 will be paid by the Marvin Company and $364,445 will be paid by the Selling Stockholders.

6.8.1

How many shares are to be sold in the primary offering? How many will be sold in the secondary offering?

a. 500,000 are to be sold in the primary offering; 400,000 are to be sold in the secondary offering.
b. 400,000 are to be sold in the primary offering; 500,000 are to be sold in the secondary offering.
c. 450,000 are to be sold in the primary offering; 450,000 are to be sold in the secondary offering.
d. 900,000 are to be sold in the primary offering; 0 are to be sold in the secondary offering
e. 765,000 are to be sold in the primary offering; 135,000 are to be sold in the secondary offering.

6.8.2

One day-post IPO, Marvin shares traded at $105. What was the degree of underpricing?

a. 7.00%
b. 7.53%
c. 17.75%
d. 31.25%
e. 41.13%

6.8.3.

One day-post IPO, Marvin shares traded at $105. What was the total underpricing costs?

a. $4,500,000
b. $5,040,000
c. $8,946,000
d. $22,500,000
e. $27,540,000

6.8.4

There are 3 kind of costs to Marvin's new issue: underwriting expense, administrative costs, and underpricing. What was the total dollar cost of Marvin issue?

a. $27,904,445
b. $27,995,555
c. $28,360,000
d. $33,035,555
e. $33,400,000

6.9. Rights issues

The Fischer Company made a rights issue at $5 a share of one new share for every four shares held. Before the issue there were 10 million shares outstanding and the share price was $6.

6.9.1

What is the maximum amount which can be raised by this rights issue?

a. $10,500,000
b. $12,500,000
c. $14,500,000
d. $16,500,000
e. $18,500,000

6.9.2

What was the prospective stock price after the issue?

a. $5.80
b. $5.95
c. $6.05
d. $6.20
e. $6.35

6.9.3

The rights issue gave the shareholder the opportunity to buy one new share for less than the market price. What is the value of the right to buy one new share?

a. $0.50
b. $0.60
c. $0.70
d. $0.80
e. $0.90

6.9.4

How far could the total value of the company fall before shareholders would be unwilling to take up their rights?

a. $20,000,000
b. $30,000,000
c. $40,000,000
d. $50,000,000
e. $60,000,000

6.10. Auctions

Which statement is false about uniform-price and discriminatory auctions?

a. In the past (more precisely, before 1998), sales of bonds by the U.S. Treasury used to take the form of discriminatory auctions rather than uniform-price auctions so that successful buyers paid their bids.
b. In a discriminatory auction, every winner is required to pay the price that she bids.
c. In 1998, the U.S. government switched to uniform-price auctions.
d. There is little cost to overbidding in a uniform-price auction, but potentially a very high cost to do so in a discriminatory auction.
e. Uniform-price auction provides less protection than discriminatory auction against the so-called “winner’s curse”.